

HRISHIKESH BELAGALI

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Objective

Driven Computer Science student with () experience seeking () opportunity

Education

Michigan State University

Expected Graduation: May 2028

Bachelor of Science in Computer Science

4.0 GPA

Bachelor of Science in Advanced Mathematics

Projects

Protein Structure Prediction | *Python, Matplotlib, NumPy, pyMOL, OVITO*

- Proposed a genetic annealing algorithm to optimize Irbäck's off-lattice model
- Generated real protein models with RMSD less than 3.
- Accurately modelled protein folding through molecular dynamics

Quadratic Unconstrained Binary Optimization Problems | *Python, Matplotlib, NumPy*

- Developed a simulated annealing algorithm using Object-Oriented Programming to determine the global minimum of QUBO problems
- Improved convergence through the implementation of dynamic stochastic tunneling.

IUPAC Name and Chemistry Retrosynthesis Generator | *Python*

- Created a database of reactions and functional groups
- Developed code to produce organic compound names in accordance with IUPAC nomenclature
- Built an algorithm for practice synthesis problems using IUPAC which closely resemble problems given in school exams
- Generated problems used by fellow classmates and underclassmen

FiveHive Survey Script | *Python, Google Forms REST API*

- Created a script with Google Forms' RESTful API to generate surveys for each of the Advanced Placement exams
- Surveys received over 1000 responses combined
- Added scripts to dynamically update questions for >30 surveys generated
- Co-led a team to analyze survey data and gain input on popular resources for over 50,000 AP students

Technical Skills

Languages: Python, JavaScript, C++

Developer Tools: VS Code, Linux WSL, SLURM

Technologies/Frameworks: Git, GitHub

Leadership / Extracurricular

12th Biennial Lake Conference

December 2020

Research

Indian Institute of Science, Bengaluru

- Carried out research on the Sarakki lake, analyzing 20 years of satellite data and tracking its pollution and eventual restoration
- Visited the lake to collect macrophytes and photos of local flora and fauna
- Review of conservation methods and proposal of a novel method to remove excess inorganic phosphates from lakes
- Presented findings at IISc, won third place at the conference

Awards

- Sahyadri Young Ecologist of the Year, IISc
- CBSE Class First in my Math Class 98%, AECS MMPS
- Best in Scientific Temper- Kumarans
- CBSE Class 12 1st in my math class- 99%, Kumarans